Zhang 2021 RCC — analysis summary

RUN 3 OF 4

GO / NO-GO GATES

Gate 1 — integration & continuity → GO

7 samples (≥369-cell threshold) on ONE connected component; clear-cell samples intermix (per-neighbourhood iLISI 3.0/7 at θ=5); the chromophobe histology forms a resolved but ADJOINING region of the same continuous manifold. Cell cycle negligible (3%). No islands.

Gate 2 — module geography → GO (non-branching)

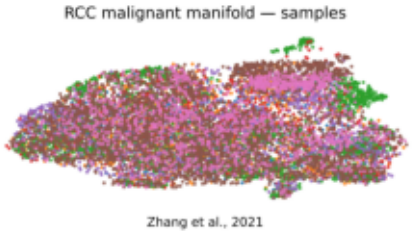
Non-branching case: a shared Gast/pSom gradient with Crest as an inverse TERMINAL gradient (pSom vs Crest r = −0.91; Crest-high decile at pseudotime 0.78). No Gast/pSom hub diverging into two competing arms. Programs present but not resolved into a bifurcation.

Gate 3 — hub cross-repression → NA

Not applicable — geometry is a single curve (non-branching), so there is no hub to test. An aggressive PPT tree over-fits the compact cloud into 33 spurious forks, not one Y.

STEP FIGURES

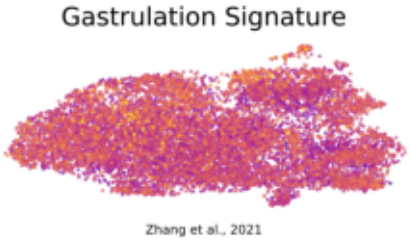
Manifold — samples



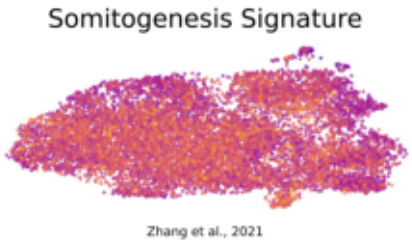
Pseudotime



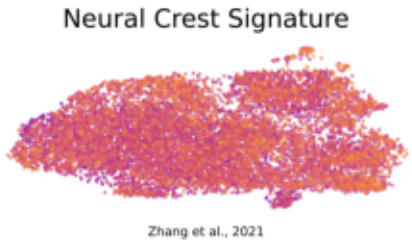
Gastrulation



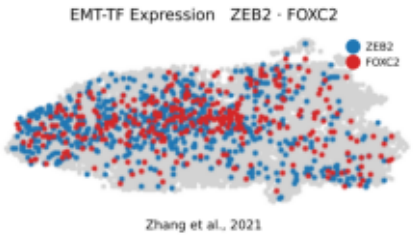
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Renal cell carcinoma (clear-cell + chromophobe) (Zhang et al., 2021), 10x — 11,218 malignant cells, 7 samples.

Verdict.
Pattern reproduces as the NON-BRANCHING case: a single gastrulation/somitogenesis axis terminating in a neural-crest gradient. No Gast/pSom bifurcation (Linear geometry). Gate 1 GO / Gate 2 GO / Gate 3 NA. Module means Gast −0.034 / pSom +0.085 / Crest −0.060. EMT-TFs ZEB2 (23.8%) + FOXC2 (12.2%).

Parameter choices

Min-cell threshold	≥369 (geom-mean of 710→192 gap, 3.7×); drops 1 small sample → 7 kept	**Integration / PCs**	Harmony θ=5, n_pcs=20 (no sc.pp.scale); iLISI-selected, per-neighbourhood
Geometry	Linear — scFates principal curve (2 tips, 0 forks, fold ratio 2.71)	**Root**	Curve tip with max Gast+pSom, min Crest (developmental origin)
Module means	Gast −0.034 / pSom +0.085 / Crest −0.060 (score_genes, union pool)	**EMT-TFs**	ZEB2 (23.8%) + FOXC2 (12.2%) — best-detected, distinct Zeb/Foxc families